# Meiosis

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Meiosis (from the Greek word, meaning 'to make smaller') is a more complicated procedure than mitosis. Both processes involve four phases: prophase, metaphase, anaphase and telophase. Meiosis undergoes the process twice and, for this reason, the phases are labelled: prophase I, metaphase I, etc. and prophase II, metaphase II, etc. to signify the stage at which the cell is at a given time.

Meiosis divides the nucleus and its contents once, then repeats the procedure a second time. By dividing the nuclear material twice, the original diploid cell is divided and reassembled as four haploid cells. Because of this division, and the 'crossing-over' of genetic material between chromatids, the four daughter cells are all genetically different.

In order to help geneticists identify them, chromosomes are numbered in every species. Humans have 46 chromosomes (23 pairs). This is written as $2n = 46$, because both parents give us 23 chromosomes, which totals 46. The numbered chromosomes form homologues.

For example: chromosome 1 will be a certain length and will contain specific genes, perhaps including one for eye colour. Its homologue, also called chromosome 1, will be the same length and contain the eye colour gene (but not necessarily coding for the same eye colour). One of the chromosomes will have been provided by the mother, the other from the father. For each of the 23 chromosomes inherited from one parent there is (or should be) a homologue. During meiosis, homologous pairs come together, including a pair of sex chromosomes.

Because humans have 23 pairs of chromosomes, the random way in which chromosomes are divided in meiosis could theoretically give up to $2^{23}$ different combinations for gametes. The 23rd pair in each set of human chromosomes is assigned by geneticists to the sex chromosomes, XX for females and XY for males.

The number of chromosomes doesn't always match up with how developed an organism is. A platypus has 52 chromosomes and a domestic dog has 78. Both have more chromosomes than humans.

1. Describe two similarities between mitosis and meiosis.
2. Describe two differences between mitosis and meiosis.
3. Rearrange the phases of meiosis in the correct order, from first to last:
   anaphase, telophase, prophase, metaphase.
4. Which type of cell division maintains the diploid number?
5. Why are chromosomes numbered?
6. 'Have the same genes (or alleles) with genetic loci arranged in the same order.' What does this describe?
7. 'Mitosis involves more steps than meiosis.' True or false?
8. 'The cells produced by meiosis are genetically identical.' True or false?
9. 'Meiosis could produce more than 8 000 000 000 possible gametes.' True or false?
10. 'Diploid cells have twice the number of chromosomes of haploid cells.' True or false?
11. 'All mammals have a diploid number of 46.' True or false?
12. 'Similar chromosomes are called homologous chromosomes.' True of false?
13. 'Possible combinations for sex chromosomes include XX, XY and YY.' True or false?
14. 'In general, the more complex an organism, the greater the number of chromosomes it has.' True or false?



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